Effect of Urbanity on Nitrate Levels in Boulder Creek

Rianna Sjostrom, Sheldon Stone, Brogan McGough, Zander Hirman

Background

- The nitrate levels in the creek affect human health but also have implications for biodiversity and aquatic life. The lower the levels of nitrate are, the better outcome for ecosystems.
- The ideal nitrate level is 10 ppm, with lower concentrates being most ideal. Highly urbanized and economic areas lead to higher nitrate levels (He, et al., 2011).
- High concentrations of nitrate can cause harmful algae blooms, leading to the depletion of oxygen. Rapid urban development and migration can lead to hotspots of riverine nitrate pollution because of the increase in urban point source discharge (Zhang et al., 2017).
- This research is important to understanding how urbanization affects nitrate levels and will help us plan future cities more efficiently and get a better grasp of how human pollution affects waterways.

Research question –

How does urbanity effect nitrate levels in Boulder Creek?

Hypothesis -

Nitrogen content in waterways will increase when surrounded by urbanity.

Prediction -

We predict that there will be higher levels of nitrate in parts of the creek that are closer to campus and downtown boulder because of increased runoff and interactions with people.

Experimental Design –

At 6 sites, 4 samples were taken randomly to reduce bias. We sampled moving water to account for confounding variables, for a total of 24 samples across the whole study. We used nitrate test strips to determine the levels of nitrate and nitrite from each site.

Dependent Variable - Nitrate Levels

Independent Variable - Urbanity (Urban or Rural see fig.1)

References -

Murphy, S. F., Verplanck, P. L., & Barber, L. B. (2003). *Comprehensive water quality of the Boulder Creek Watershed, Colorado, during high-flow and low-flow conditions, 2000* (USGS Water-Resources Investigations Report 03–4045). U.S. Geological Survey. https://pubs.usgs.gov/publication/wri034045

He, B., Kanae, S., Oki, T., Hirabayashi, Y., Yamashiki, Y., & Takara, K. (2011). Assessment of global nitrate pollution in rivers using an integrated biogeochemical modeling framework. *Water Research, 45*(8), 2573-2586. https://doi.org/10.1016/j.watres.2011.02.011

Zhang, X., Wu, Y., & Gu, B. (2017). Influence of rapid rural-urban population migration on riverine nitrate pollution: Perspective from ammonia-nitrate. *Environmental Science and Pollution Research, 24*(10), 9223–9231. https://doi.org/10.1007/s11356-017-0322-6

Lopitz, J., Linenfelser, B., Owen, C., Widner, J., Ariniello, A., Elkins, J., Wertz, R., Fowlkes, C., Forrester, M., Mays, J., Palmer, W., Gaioni, J., Marko, K., Youngberg, C., Rees, D., Clary, J., Behling, K. (2016), Boulder Creek and St. Vrain Creek Watersheds Annual Water Quality Analysis for 2015, https://www.keepitcleanpartnership.org/wp-content/uploads/2018/09/St.VrainBoulderWQAnalysis_FinalAugust2016.pdf

Results

We used a linear regression model to analyze our data and used box plots to present it. Our box plots show that there is an effect of urbanity on nitrate levels, as you get closer to the city the levels of nitrate increase. The predictor variable would be urbanization, and the response variable would be the nitrate concentration for both of the following graphs.

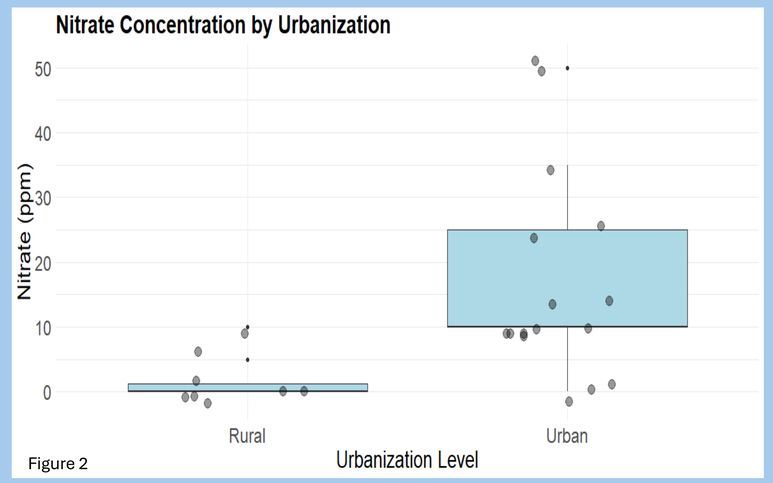


Figure 2

This effect size plot shows the estimated increase in nitrate due to urbanization. With no overlap, it shows that there is a clear effect on nitrate levels by urbanization

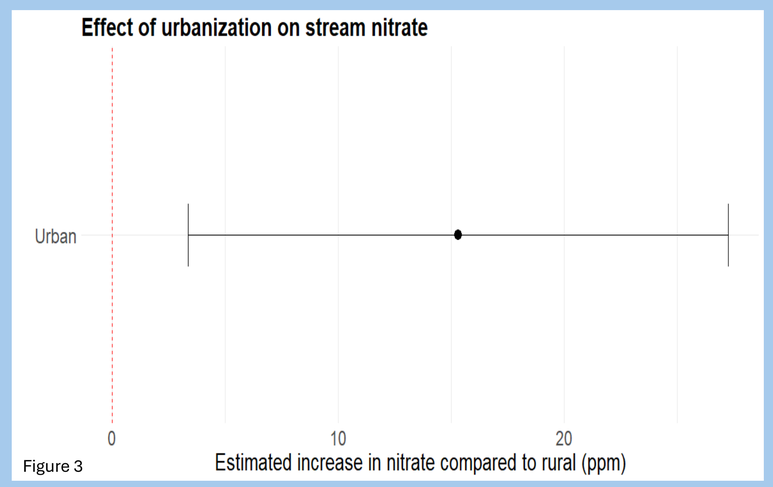


Figure 3

Contributions -

Zander- Figure 1, Figure 2, Figure 3, data collection
Sheldon- Background, Discussion, figure headings, Data Collection
Brogan- Methods (Question, Hypothesis, Prediction), Experimental Design, Discussion Context, References, Data Collection
Rianna- Discussion, Background, Data collection and analysis

Discussion

- Our data shows an increase in the levels of nitrate as we got closer to the center of Boulder.
- In the box plot it shows a clear effect on nitrogen by urbanity. The effect size plot shows an obvious effect of urbanity on nitrogen, proving our hypothesis correct.
- Data showed that there was an average of 2.5 ppm of nitrate in mountain areas and 21.66ppm in urban areas. This shows that human activity around the creek increased nitrate levels.
- Confounding variables effecting our data include how recently it rained, seasonal change in vegetation, water speed, and elevation. Almost all urban areas in Boulder are at low elevation, so elevation change is unavoidable.
- One solution is to study over a span of time. Gathering data in different seasons as well as accounting for periods of rainfall can help accommodate these variables.
- Our results are consistent with a USGS Study that found that nitrogen levels increase in urban areas due to Boulder Wastewater Treatment Plant runoff, however they reported numbers below 2ppm. (USGS, 2003)
- Our study deviates from a study from Wright Water Engineers which found that levels of nitrate increased in urbanized areas, however levels of nitrate in that study are well below 10ppm, while we found an average of 21.66 ppm in urban areas. (WWE, 2016)
- This research can help scientists and the people of Boulder better understand the health of the creek at many different points along it. Additionally, this research could be applied to different cities to compare the relative nitrogen levels in their creek water.

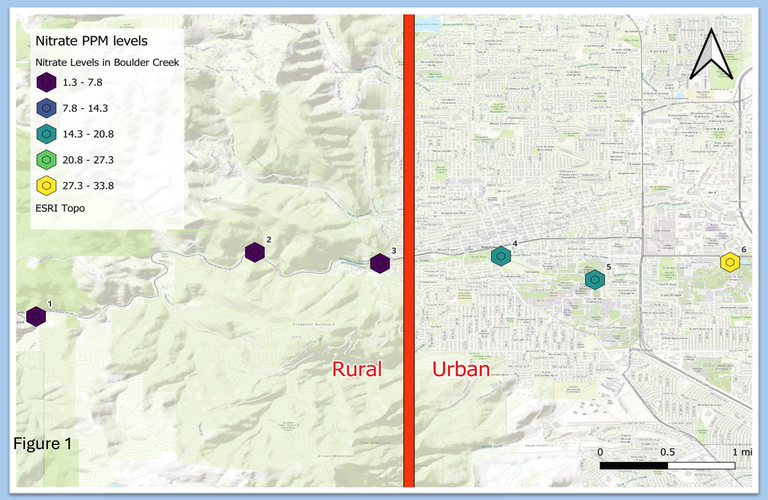


Figure 1

Points show sample sites, and the color key shows average nitrogen concentration at each point